

Digital Forensics Laboratory Report

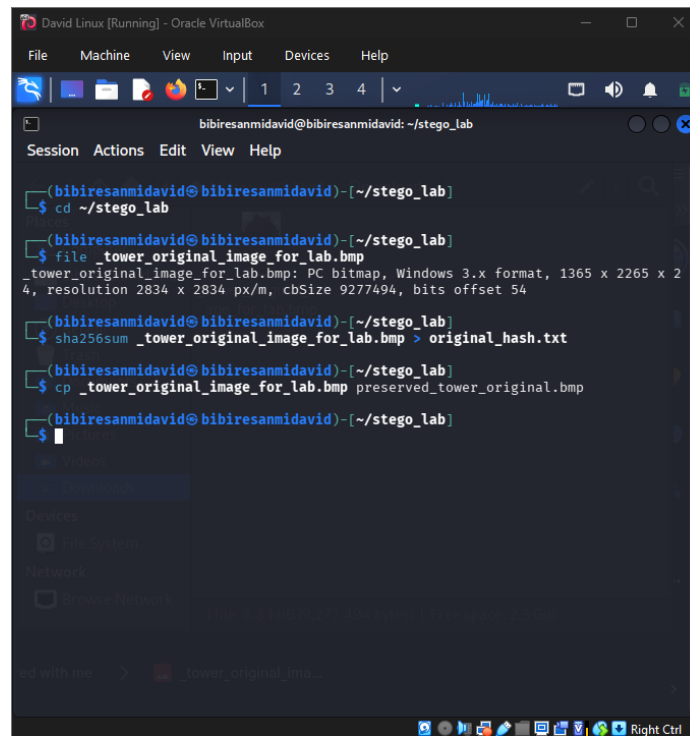
Module: ICDFA Module 8 – Steganography Practical

Investigator: Bibiresanmi David Oluwagbenga

Part A: Evidence Preparation and Preservation

The initial carrier image was verified, and a baseline cryptographic hash was established to maintain evidence integrity before any modifications occurred.

- **Evidence File:** _tower_original_image_for_lab.bmp
- **File Type:** PC bitmap, Windows 3.x format, 1365 x 2265 x 24
- **File Size:** 9,277,494 bytes
- **Preservation:** Copied to preserved_tower_original.bmp to ensure the original evidence remained untampered.



```
David Linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

bibiresanmidavid@bibiresanmidavid: ~/stego_lab
Session Actions Edit View Help

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$ cd ~/stego_lab

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$ file _tower_original_image_for_lab.bmp
_tower_original_image_for_lab.bmp: PC bitmap, Windows 3.x format, 1365 x 2265 x 24, resolution 2834 x 2834 px/m, cbSize 9277494, bits offset 54

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$ sha256sum _tower_original_image_for_lab.bmp > original_hash.txt

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$ cp _tower_original_image_for_lab.bmp preserved_tower_original.bmp

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$
```

Figure A1 – Carrier image verification, hashing, and preservation.

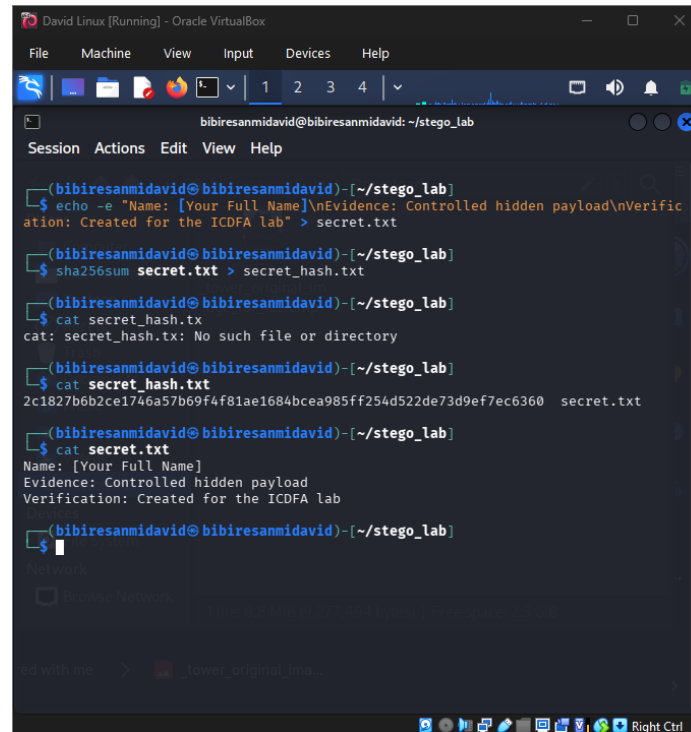
Part B: Secret Evidence File Creation

A controlled payload file (secret.txt) was generated containing specific identifying information for the forensic exercise.

File Content:

Name: Bibiresanmi David Oluwagbenga
Evidence: Controlled hidden payload
Verification: Created for the ICDFA lab

- **SHA-256 Hash:** 2c1827b6b2ce1746a57b69f4f81ae1684bcea985ff254d522de73d9ef7ec6360



```

David Linux (Running) - Oracle VirtualBox
File Machine View Input Devices Help

bibiresanmidavid@bibiresanmidavid: ~/stego_lab
Session Actions Edit View Help

(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
$ echo -e "Name: [Your Full Name]\nEvidence: Controlled hidden payload\nVerification: Created for the ICDA lab" > secret.txt

(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
$ sha256sum secret.txt > secret_hash.txt

(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
$ cat secret_hash.tx
cat: secret_hash.tx: No such file or directory

(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
$ cat secret_hash.txt
2c1827b6b2ce1746a57b69f4f81ae1684bcea985ff254d522de73d9ef7ec6360 secret.txt

(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
$ cat secret.txt
Name: [Your Full Name]
Evidence: Controlled hidden payload
Verification: Created for the ICDA lab

(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
$

```

Figure B1 – Creation and hashing of the secret evidence file.

Part C: Embedding the Hidden File

The controlled payload was concealed within the carrier image using the Steghide utility and the authorized lab password (1234).

- **Target File:** tower_stego_lab4.bmp
- **Stego Image SHA-256 Hash:**
d9e072a6e150266cfe3c70a9d022313775070ac66669ac7cab47e2fdc6bd9a0e

```
David Linux (Running) - Oracle VirtualBox
File Machine View Input Devices Help

bibiresanmidavid@bibiresanmidavid: ~/stego_lab

Session Actions Edit View Help

Unpacking libmcrypt4:amd64 (2.5.8-8+b2)...
Selecting previously unselected package steghide.
Preparing to unpack ../steghide_0.5.1+git20240220-2_amd64.deb...
Unpacking steghide (0.5.1+git20240220-2)...
Setting up libmcrypt4:amd64 (2.5.8-8+b2)...
Setting up steghide (0.5.1+git20240220-2)...
Processing triggers for libc-bin (2.42-16)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.2.6)...

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$ steghide embed -ef secret.txt -cf tower_original_image_for_lab.bmp -sf tower_stego_lab4.bmp -p 1234
embedding "secret.txt" in "tower_original_image_for_lab.bmp" ... done
writing stego file "tower_stego_lab4.bmp" ... done

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$ sha256sum tower_stego_lab4.bmp > stego_hash.txt

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$ ls
original_hash.txt          stego_hash.txt
preserved_tower_original.bmp tower_original_image_for_lab.bmp
secret_hash.txt            tower_stego_lab4.bmp
secret.txt

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$ cat stego_hash.txt
d9e072a6e150266cfe3c70a9d022313775070ac66669ac7cab47e2fdc6bd9a0e tower_stego_lab4.bmp

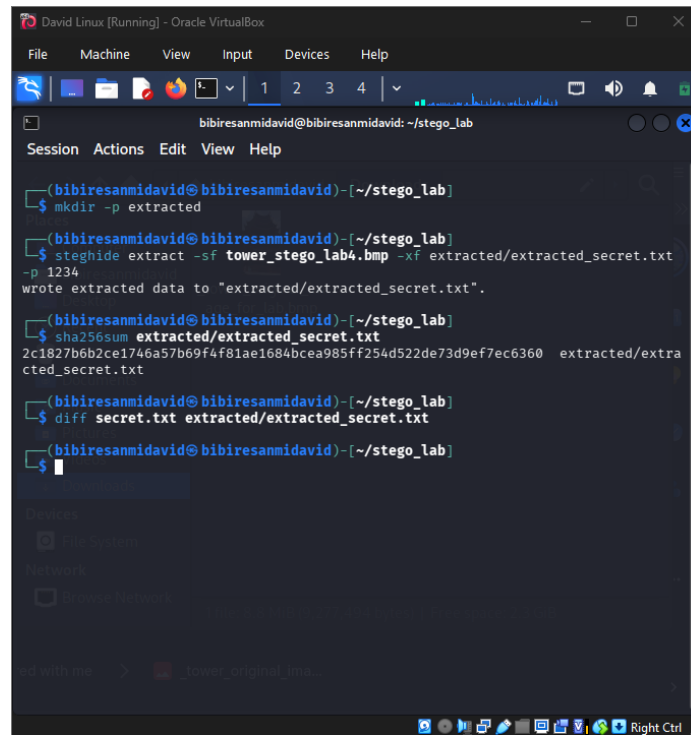
(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$
```

Figure C1 – Embedding secret.txt into the carrier image with Steghide.

Part D: Extraction and Verification

To prove data recoverability, the payload was extracted from the stego image into a dedicated directory using the known password.

- **Extracted File Hash:** 2c1827b6b2ce1746a57b69f4f81ae1684bcea985ff254d522de73d9ef7ec6360
- **Verification:** The diff command returned no output, confirming the extracted file is a mathematically perfect match to the original secret.txt.



```
David Linux (Running) - Oracle VirtualBox
File Machine View Input Devices Help

bibiresanmidavid@bibiresanmidavid: ~/stego_lab
Session Actions Edit View Help

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$ mkdir -p extracted

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$ steghide extract -sf tower_stego_lab4.bmp -xf extracted/extracted_secret.txt
-p 1234
wrote extracted data to "extracted/extracted_secret.txt".

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$ sha256sum extracted/extracted_secret.txt
2c1827b6b2ce1746a57b69f4f81ae1684bcea985ff254d522de73d9ef7ec6360  extracted/extr
acted_secret.txt

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$ diff secret.txt extracted/extracted_secret.txt

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$
```

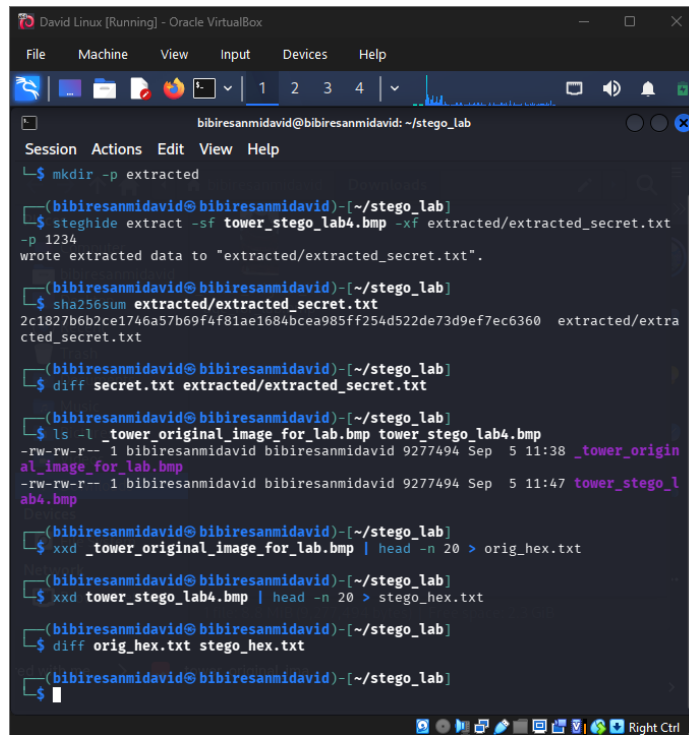
Figure D1 – Extraction of the hidden payload and hash verification.

Part E: Structural Analysis (Original vs. Stego)

A comparative analysis was conducted between the clean carrier and the embedded stego file.

Metric	Original Image (tower_original_image_for_lab.bmp)	Stego Image (tower_stego_lab4.bmp)
File Size	9,277,494 bytes	9,277,494 bytes
SHA-256 Hash	[Original hash documented in orig_hex.txt]	d9e072a6e150266cfe3c70a9d02231377507 0 ac66669ac7cab47e2fdc6bd9a0e
Visual Appearance	Identical	Identical

Theoretical Explanation: Steghide uses Least Significant Bit (LSB) substitution, replacing the least important bits of the carrier's pixel data with the secret file's binary data. This completely alters the underlying binary structure—causing the cryptographic hash to change—but results in visual shifts too minuscule for the human eye to detect.



```
David Linux (Running) - Oracle VirtualBox
File Machine View Input Devices Help

bibiresanmidavid@bibiresanmidavid: ~/stego_lab
Session Actions Edit View Help
└─$ mkdir -p extracted

(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
└─$ steghide extract -sf tower_stego_lab4.bmp -xf extracted/extracted_secret.txt
-p 1234
wrote extracted data to "extracted/extracted_secret.txt".

(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
└─$ sha256sum extracted/extracted_secret.txt
2c1827b6b2ce1746a57b69f4f81ae1684bcea985ff254d522de73d9ef7ec6360  extracted/extrac
cted_secret.txt

(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
└─$ diff secret.txt extracted/extracted_secret.txt

(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
└─$ ls -l _tower_original_image_for_lab.bmp tower_stego_lab4.bmp
-rw-rw-r-- 1 bibiresanmidavid bibiresanmidavid 9277494 Sep  5 11:38 _tower_origin
al_image_for_lab.bmp
-rw-rw-r-- 1 bibiresanmidavid bibiresanmidavid 9277494 Sep  5 11:47 tower_stego_l
ab4.bmp

(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
└─$ xxd _tower_original_image_for_lab.bmp | head -n 20 > orig_hex.txt

(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
└─$ xxd tower_stego_lab4.bmp | head -n 20 > stego_hex.txt

(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
└─$ diff orig_hex.txt stego_hex.txt

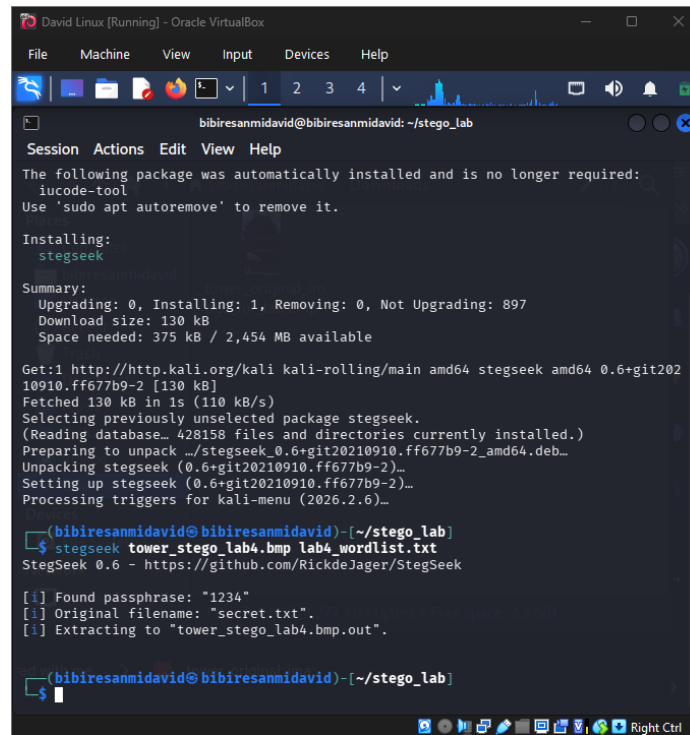
(bibiresanmidavid@bibiresanmidavid)-[~/stego_lab]
└─$
```

Figure E1 – Byte-level (xxd) and hash comparison of original vs. stego image.

Part F: Controlled Password Dictionary Test

A brute-force password recovery workflow was successfully executed against the stego image.

- **Tool Used:** StegSeek 0.6
- **Target Image:** tower_stego_lab4.bmp
- **Result:** Passphrase "1234" successfully found. Original filename "secret.txt" extracted to "tower_stego_lab4.bmp.out".



```
David Linux (Running) - Oracle VirtualBox
File Machine View Input Devices Help

bibiresanmidavid@bibiresanmidavid: ~/stego_lab

Session Actions Edit View Help

The following package was automatically installed and is no longer required:
iucode-tool
Use 'sudo apt autoremove' to remove it.

Installing:
stegseek

Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 897
Download size: 130 kB
Space needed: 375 kB / 2,454 MB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 stegseek amd64 0.6+git20210910.ff677b9-2 [130 kB]
Fetched 130 kB in 1s (110 kB/s)
Selecting previously unselected package stegseek.
(Reading database... 428158 files and directories currently installed.)
Preparing to unpack ./stegseek_0.6+git20210910.ff677b9-2_amd64.deb...
Unpacking stegseek (0.6+git20210910.ff677b9-2)...
Setting up stegseek (0.6+git20210910.ff677b9-2)...
Processing triggers for kali-menu (2026.2.6)...

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$ stegseek tower_stego_lab4.bmp lab4 wordlist.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: "1234"
[i] Original filename: "secret.txt".
[i] Extracting to "tower_stego_lab4.bmp.out".

(bibiresanmidavid@bibiresanmidavid)~[/stego_lab]
$
```

Figure F1 – StegSeek dictionary attack recovering the embedded payload.

Part G: Independent Student Task & Theoretical Conclusions

Steganography vs. Encryption: Encryption scrambles a message to hide its meaning (making it unreadable without a key), whereas steganography hides the actual existence of the message.

Forensic Importance: Bad actors use steganography to bypass Data Loss Prevention (DLP) systems, exfiltrate stolen data, or store illicit materials in plain sight to evade investigators scanning for encrypted volumes.